Explainable Financial Alerts for Retail Investors

Integrating Statistical Anomaly Detection and News–Market Impact Correlation

Henrique J. S. Santos
Advisor: Prof. Luís Gomes   Co-advisor: Rafael Silva

MEIA, Instituto Superior de Engenharia do Porto

Defence

The problem

- Most Americans own stock; many now trade through commission-free apps.
- The US equity market reached **US$62.2 trillion** (end 2024).
- Retail investors face *continuous* price moves and news, **without** the tools of banks and funds.
- The AI spreading through finance is largely **opaque**, a poor fit for someone who must act and live with the result.

The three questions

When a holding moves, everyone asks the same three:

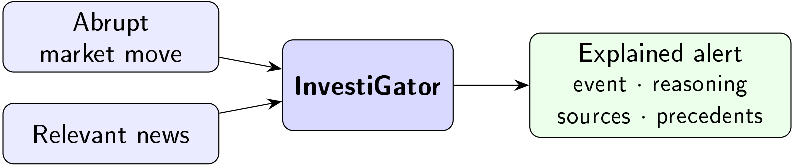
1. Is this *unusual for this stock*?
2. Is it the *company*, or the market?
3. Has this *happened before*, and what followed?

Free tools answer **none** of the three; they report the percentage, which is where question 1 *starts*. A Bloomberg terminal answers all three, for about $2,000 a month. The gap is not data, which is abundant and free. It is **context you can check**.

What InvestiGator does (and does not do)

It does *not* predict prices. It explains.

For each abrupt move or relevant news item, it shows *what* was detected, *how*, the *sources*, and which *past events* resemble it, with what happened to prices afterwards.



Research questions

RQ1 Can **transparent** statistical methods detect abrupt moves consistently enough to be useful, while staying explainable?

RQ2 Can a news–market engine retrieve **genuinely analogous** precedents and quantify their impact **without lookahead**, better than trivial baselines?

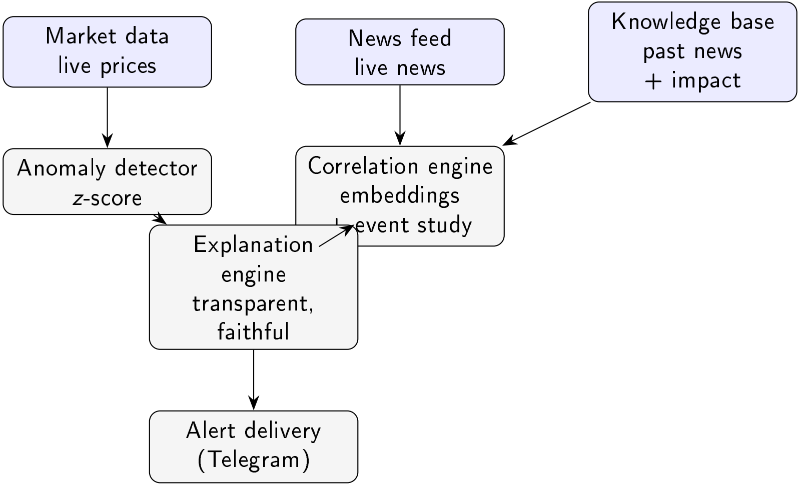
RQ3 Can the explanations be **faithful** to what the system computes, and useful to a retail investor?

RQ4 Can a **supervised model**, trained on historical news and market context, usefully **prioritise** which news deserve an alert, beyond volatility baselines — **never** predicting direction?

Contribution: Artificial Intelligence *Engineering*

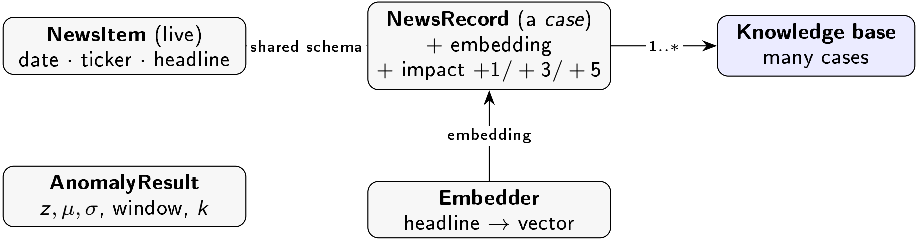
Not a new algorithm: the **disciplined integration, application and critical evaluation** of existing components — plus **one model trained by the author** (materiality triage) — into a working, explainable, reproducible system.

Architecture



Single-responsibility components, common schema; two triggers converge on one explanation step.

The data model — the objects



A *case* = one past headline + what happened to its price. Live **NewsItem** and historical **NewsRecord** share the schema, so a new headline is directly comparable.

The data, at every stage — one real headline

1 • RAW
retrieved
`2018-05-10`
`NVDA`
`"Nvidia Blows Out Q1..."`
+ daily prices

→

2 • CLEAN
no lookahead
align to 1st trading day
prices → log-returns
drop live summary

→

3 • REPRESENT AI
model's view
text → SBERT **384-d** vector
context: vol20, mom5, ret
label: material? **yes**

→

4 • MEASURE
two questions
triage → p(material)
retrieval → precedents
PR-AUC · prec@k

Every value is genuine (Nvidia, 10 May 2018). The "AI" is the **REPRESENT** step: text becomes a 384-dimensional embedding, the event becomes context features, the outcome becomes a label.

Market trigger: transparent by design

Rolling *z*-score, no lookahead:

$$z_t = \frac{r_t - \mu_t}{\sigma_t}, \quad \text{flag if } |z_t| > k$$

μ_t, σ_t over the window *before* t. Every quantity is exposed to the explanation.

Worked example (real)

TSLA, 24 Oct 2024 (post-earnings):
$\mu = -0.92\%$, $\sigma = 2.73\%$, $r = +19.8\%$
$\Rightarrow z = +7.61$ (flagged at $k = 3$).

A fixed % threshold would treat a calm and a volatile stock alike; the *z*-score does not.

Question 2: is it the company, or the market?

A two-factor market model, estimated **only on data before the day being explained**:

$$r_t = \alpha + \beta_m r_t^m + \beta_s \tilde{r}_t^s + \varepsilon_t$$

The sector factor is **orthogonalised** against the market (an ETF and the index are collinear; without it the attribution is meaningless).

Betas are **shrunk by precision**, not clipped: a noisy 20-day estimate is pulled toward its prior, a clean one is left alone.

Real, from the live system

AMZN -1.84% **today**

$= -1.66\%$ market $\cdot$ -0.37% sector $\cdot$ $+0.19\%$ **company**

"Moved with the whole market."

The stock fell almost two percent and the company's own contribution was **positive**. For the long-term holder, the most valuable output a system can give is often **permission to ignore something**.

Components sum to the observed move by construction, so the line cannot lie.

What it watches — and why two of the twelve are not tech

XLK — technology

Alphabet

XLF JPMC XLE ExxonMobil XLV J&J

Until August 2026 the watchlist was **ten**, and **nine of them sat in XLK**. So the question "was it the sector?" nearly always came back with the same answer — not because that was the answer, but because the watchlist gave the sector factor nothing to vary against. **XOM (energy) and JNJ (healthcare) were added to fix that**, and the effect showed up immediately in live output: **XOM** -0.98% **on a day its sector was** $+0.93\%$ — the sector pulling the *other* way, which nine technology names could not have shown.

This is a change to the *evaluation*, not to the layout.

News trigger: retrieve analogous precedents, measure impact

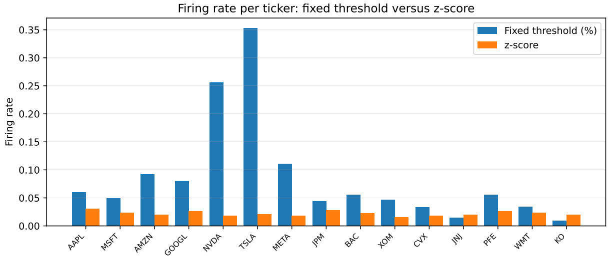
1. Embed the headline (Sentence-BERT).
2. Cosine similarity vs the knowledge base.
3. Return top-k precedents.
4. Measure their impact at $+1/+3/+5$ days, from the event-day close (**no lookahead**).

Case-based reasoning: evidence, not a forecast.

Worked example (real)

Query: "Nvidia demand surges on AI chip orders"
$\rightarrow$ NVDA guidance (sim 0.60) +5d +3.5%
$\rightarrow$ MSFT cloud-AI (sim 0.38) +5d +10.9%
$\rightarrow$ NVDA accelerator (sim 0.38) +5d +4.9%
Cross-ticker match (MSFT); mean +5d $\approx$ +6.5%.

Result 1: anomaly detection is consistent across the market

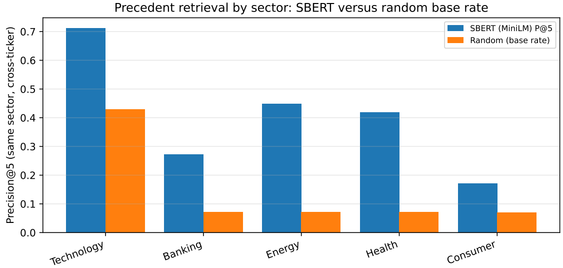


- Firing-rate range across 15 tickers: **0.015** (z-score) vs **0.344** (fixed %).
- $> 20\times$ more consistent, a **label-free** argument.
- Vs extreme-move proxy: $F_1 = 0.516$ vs 0.218.
- Window ablation: F_1 0.385 / 0.516 / 0.678 (10/20/60 d).

Result 2: semantic retrieval beats every baseline

Method	P@5	P@10
SBERT (MiniLM)	0.514	0.478
SBERT (MPNet)	0.538	0.506
Lexical	0.346	0.314
Recency	0.126	0.106
Random	0.240	0.240

Cross-ticker, 3,714 headlines, 5 seeds ($\pm\sim 0.01$).



Lift largest where vocabulary is distinctive (energy, health).

Validated at scale on multi-year FNSPID (~80k headlines): **P@5** 0.595; direction consistency 0.71 sits at the 0.69 chance floor (theme, not direction).

Result 3: explanations are faithful by construction

- Each alert is rendered *directly* from the computed objects.
- So the text **cannot diverge** from the computation.
- Checked by an automated test (alert reproduces each precedent's date/ticker/impact).
- Every alert ends with a no-prediction disclaimer.

Real alert (news trigger)

```
News alert -- NVDA
“Nvidia raises guidance...AI data-centre”
Avg 1-day move:  −1.97% (over 5; 3 shown)
- MSFT (0.68) +1d −3.46%
- META (0.68) +1d −2.65%
- GOOGL (0.64) +1d −0.46%
observed past outcome, not a prediction
```

Result 3b: the same guarantee, for generated language

A language model was added **strictly as a narrator**: it may phrase the evidence, never supply it. A guard validates every response before delivery and **discards the whole thing** on:

- a number absent from the evidence,
- a figure whose **sign** was inverted,
- predictive or advisory wording,
- an attribution contradicting the computed one.

The guarantee is a property of **the system**, not of the model behaving well.

Two numbers, deliberately

Pre-guard: the raw model violated the evidence in a few cases.
$\Rightarrow$ *why the guard exists.*

Delivered: **zero** violations.
$\Rightarrow$ *the guard is what is on trial.*

Honest caveat: the second is partly circular, since the same checker decides and grades. The counterweight is a red-team corpus: an independent exercise broke the *first* design (a blocklist) with 29 reproduced bypasses. Paraphrase space is unbounded and any list of it is finite, so the guard was rebuilt as a **closed vocabulary**. Those attacks are now permanent regression tests.

Result 4: learned triage — trained, calibrated, honest

- **Task:** P(abnormal move follows news) — materiality, **never** direction. Labels from our own event study ($|\text{ticker} - \text{SPY}| \geq 2\%$ over $(d, d{+}3]$).
- **Corpus:** FNSPID 2018–2023, 79,753 examples; temporal split by day + embargo; calibration on validation only.
- **Anti-lookahead:** a unit test *mutates future prices* — features unchanged, label changes.

Model	PR-AUC	P@5/day
Always (floor)	0.378	0.163
Volatility LR	**0.542**	**0.632**
Context LR	0.538	0.632
Context+text LR	0.496	0.585
Gradient boosting	0.469	0.551

The honest verdict (pre-committed)

Within a 5-alerts/day budget, triage nearly **quadruples precision** (0.632 vs 0.163) *on held-out data from the training corpus* — but **no text model beat volatility**. A follow-up ablation added *five* more cheap context features (market regime, momentum, standardised reaction, downside risk): PR-AUC 0.535, again **no gain**. A *fair* re-test (tuned C, PCA-reduced text, a FinBERT domain encoder) still never beat volatility, so the negative is **robust**, not an under-fit artefact. The signal is market context; reported as it fell.

The bug that showed the model was never deciding

What happened. Nvidia rose sharply on a supply announcement. The channel carried **two lesser stories** about Nvidia that day and **never carried that one**.

Why. The daily cap of two alerts per company was served **in arrival order**. Two immaterial morning stories exhausted the budget; the afternoon story was dropped **without a trace in any log**.

The uncomfortable part

The instrument was already there. Triage exists to rank by materiality, and lifts precision from 0.163 to 0.632 inside a 5-alert budget — and **the cap never consulted it**. The score was measured, then ignored.

The first fix did not work, and that is the interesting part. Sorting the cap by score can only reorder candidates present *together*, and the scan emits one headline per company per cycle — so two stories competing for the same quota never coexist. Across a day the order stayed arrival order. What governs the cap is an **escalating floor**: the k-th alert of the day must clear a higher bar (0.49, then 0.64), both *derived* from the cost sweep, not chosen.

And the fix had its own bug. Comparing *rendered alerts* to catch repeated stories collides on shared template

Result 5: what the system does *not* know

Four studies that examine the system from outside. Each ends in a decision **not** to change production, taken on the evidence. The fourth, multi-signal convergence, adapted from **worldmonitor.app** on the co-supervisor's recommendation: fusing four independent signals beat the best single one at *one* alert budget out of three, so it stays a measurement. Its by-product, a **volume** detector at zero data cost, is kept.

Event types

Clustered the embeddings against a rubric written *before* clustering. Agreement (chance-corrected): event **0.358** > ticker 0.188 > sector 0.130.

So the space *does* encode event type, more than subject. But separation is weak (silhouette 0.084) $\Rightarrow$ **not** wired to retrieval: a wrong event type drops valid precedents silently.

Conformal guarantee

Distribution-free coverage on the *frozen* model (PR-AUC reproduced exactly first).

Holds at 90%/80%; **breaks at 95%** under a temporal split. Tighter coverage leans on the tail, and the tail moves first.

The price: to promise 90%, a definite call on only **39.5%** of headlines.

Drift, measured

The limitation we kept *asserting*, finally sized.

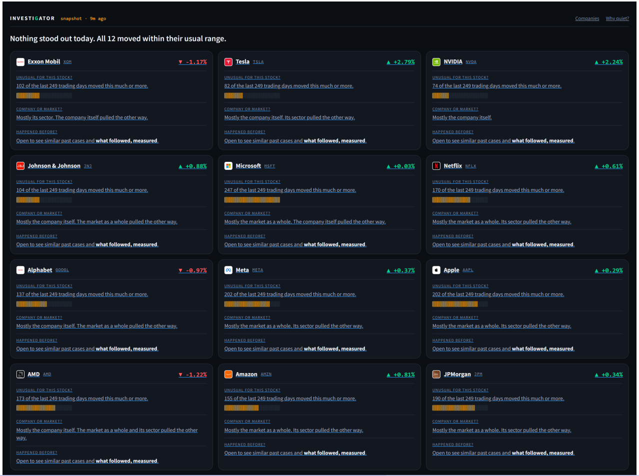
Pre-event volatility PSI **0.281** (significant); returns features 0.020 / 0.014 (stable).

Label prevalence *oscillates* (0.385 $\rightarrow$ 0.470 $\rightarrow$ 0.378), it does not trend. Gives a **retraining trigger** instead of an intuition.

Why this matters more than another positive number

The 39.5% figure **explains** Result 4 from an independent angle, without training anything new: the

The product, live



A genuine screenshot, not a mockup. **Every card opens with a sentence, then the numbers behind it.**

- **Rarity is a count, not a probability** — "only 5 of the last 249 days moved this much". Turning z into a probability needs normality; returns have fat tails, so it would fail on exactly the days that matter.
- **Quiet is evidenced, not asserted** — a calm day shows the count that makes it calm.
- **When the two measures disagree, both are shown** — Microsoft: below the alert threshold against its own 20 days, yet only 5 of 249 moved as much.

Built with — free tools and open data, end to end

Data sources & APIs

FNSPID (Hugging Face) · yfinance · Finnhub · RSS feeds · SPY (market proxy)

Machine learning

Sentence-BERT / MiniLM · scikit-learn · PyTorch (CPU) · Platt calibration · ONNX Runtime

Product & delivery

Telegram Bot API · Streamlit · Plotly

Engineering & infrastructure

Python 3.12 · GitHub Actions (cron) · pytest + ruff · joblib

No paid APIs, no GPUs, no always-on server — reproducible on a laptop.

Result 9: the gate was measured in production, and it did not transfer

Every triage decision is logged and, days later, labelled with what the price actually did — same rule as training. Over **530** matured decisions:

Alerts the gate...	were material
kept ($p \geq 0.5$)	0.592
suppressed ($p < 0.5$)	0.647

The gate is on the *wrong* side, though not significantly ($z = -1.28$, $p = 0.20$).

Which failure is it?

Two failures look identical here and need opposite fixes. If the score still **ranks**, recalibrate (two parameters). If it does not rank, recalibration is useless — a sigmoid is **monotone**, it preserves the order exactly.

Measured: **ROC-AUC** 0.494, cluster CI $[0.391, 0.601]$ over 145 (ticker, day) units. Centred on chance. **No order to preserve.**

Why, and it is not a broken model — it is a redundant one. Materiality among logged decisions runs at 0.626 against 0.378 in training, because decisions are only logged for headlines that *already* cleared the relevance and freshness filters. The cheap upstream stages had removed what the learned stage was trained to remove. **A model evaluated in isolation and deployed behind filters was never evaluated on the distribution it would see.** The gate stays on as a *volume* control (it suppresses $\sim$3/5) and because it is the instrumented part — switching it off would end the measurement that found this.

Limitations (stated plainly)

- Relevance is a **same-sector proxy**, not a human judgement.
- Retrieval corpus is a **recent** free-tier window (since validated at scale on FNSPID, P@5 0.595); the *triage* study did use multi-year FNSPID (14/15 tickers — Meta is "FB" in the corpus).
- Displayed precedent impact is a **raw** post-event return (triage labels *are* market-adjusted).
- Similarity is **thematic, not directional**: a cluster can mix up/down moves, so the mean is theme-level *evidence*, not a forecast. We *tested* the obvious fix (filter by event type) and it does not hold up — see Result 5.
- **Confidence is thin, and now quantified:** guaranteeing 90% coverage leaves a definite call on only **39.5%** of headlines.
- **Train/deploy drift is measured, not just claimed:** concentrated in volatility (PSI 0.281), and cyclical rather than trending.
- **No human usefulness study** yet (RQ3 usefulness open).
- By design: **no price prediction, no trading**, so profit metrics are out of scope.

All results are reproducible from versioned scripts (re-run: identical numbers). Two of the limitations above used to be assertions; they are now measurements.

Conclusions

- **RQ1 (yes):** transparent z-score detects consistently across the market and explains every flag.
- **RQ2 (yes, validated at scale):** semantic retrieval far above baselines (P@5 0.595 on multi-year FNSPID); impact measured without lookahead.
- **RQ3 (faithful yes; useful open):** explanations cannot diverge from the computation; human usefulness is future work.
- **RQ4 (no on text; mechanism yes *offline*):** triage $\approx 4\times$ within-budget precision on held-out data, but volatility stayed unbeaten — the *second* fair "learned vs simple" test the transparent choice won (first: Isolation Forest vs z-score, F_1 0.271 vs 0.530). Measured in production the same gate ranks at chance (Result 9), so the mechanism claim is an evaluation-set claim and is stated as one.

Contribution

A working, **explainable, reproducible** financial-alert system for retail investors — existing components plus one author-trained triage model, engineered into a transparent whole, with an honest evaluation.

Thank you

InvestiGator

Explainable financial alerts for retail investors

Questions welcome.

Anticipated questions

- **Why not predict prices?** Market efficiency (Fama 1970): public news is impounded fast; we *measure* reactions, honestly.
- **Why *z*-score, not Isolation Forest/GARCH?** Transparency for a low-dimensional signal — and we *tested* it: a causal Isolation Forest on the same information lost (F_1 0.271 vs 0.530).
- **Your trained model lost to the volatility baseline. Failure?** No: the comparison was *pre-committed* and answers RQ4 with evidence. Triage gives $\approx 4\times$ within-budget precision on held-out data, and the deployed variant uses exactly the features that carry the signal.
- **So does the ML work in production?** *As a selector, no, and I measured it* (ROC-AUC 0.494). As a volume control, yes. The reason is instructive: the upstream filters already removed most of what it was trained to remove, so it is redundant rather than broken — a lesson about placing a learned component *inside* a pipeline.
- **Does the system learn automatically?** No. It does *inference* with fixed weights and *collects labels*; retraining is possible and documented but has never run. Calling that continual learning would be an overclaim.
- **How do you know the triage model never sees the future?** A unit test mutates post-event prices and asserts features are unchanged while the label changes; temporal split by unique day with a five-day embargo.
- **Why precision@*k* and a sector proxy?** Standard ranked-retrieval metric; objective, reproducible; human

Anticipated questions (continued)

- **Isn't "zero delivered violations" circular, since your own checker grades it?** Partly, and the thesis says so. That is why it is not the only test: an independent red team broke the *previous* guard with 29 reproduced bypasses, built without sight of the current design, and those attacks are now regression tests.
- **So you built a multi-agent system?** No, and I will not claim it. One model calling a few functions is not multi-agent. The narrator is a **pure function** from evidence to a paragraph.
- **Why exclude analyst price targets? They are free.** Because importing someone else's forecast into a system defined by not forecasting is a contradiction. Coherence was preferred to coverage, and it is stated as a position in Chapter 6.
- **Why no portfolio features?** Holdings are personal financial data, and "your top positions are one bet" is advice however it is phrased. That crosses the boundary the ethical claim depends on.
- **Where does the 0.5 triage threshold come from?** Originally nowhere, which was the honest problem. Sweeping it under an explicit cost ratio showed it implicitly assumed a miss costs about the same as a false alarm; under equal costs the optimum sits where the constant already was. Vindicated, but now *derived*.
- **Your alerts arrive hours late.** Measured: a median of about three hours, and the cause is the free scheduler, not the method. The always-on path is written and costs about a minute; deploying it was out of scope for the evaluation, and the honest number is displayed in the product rather than hidden.